

Course: B. Tech.

Branch: Civil Engineering/Civil and infra

Semester: V

Subject Code & Name: BTCVC501/BTCIC501 & Design of Steel Structures

Max Marks: 60

Date: 01/01/2024

Duration: 3 Hrs.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.
5. Use of IS 800-1984, IS 800-2007, IS 875-1987 and steel table is permitted.

(Level/CO) Marks

Q. 1 Solve Any Two of the following.		12
A) Differentiate between WSM and LSM	Remember	6
B) Design a connection to connect two plates 80 mm x 8 mm and 100 mm x 10 mm are connected in a lap joint to transmit pull equal to the full strength of the plate using bolted connection.	(L3/CO2)	6
C) An angle ISA 50 × 50 × 6 mm is to carry an axial load of 85 kN. The angle is to be connected to 10 mm thick gusset plate. Design welded connection.	(L3/CO2)	6
Q.2 Solve Any One of the following (A or B).		12
A)		6
1) 2ISA 60 × 60 × 8 mm connected to gusset plate 10 mm thick with 14 mm dia. bolts. Find the permissible strength in axial tension. Take $f_y = 250\text{MPa}$. Case A: 2 ISA connected same side of gusset plate. Case B: 2 ISA connected either side of gusset plate.	(L3/CO3)	
2) Design double angle discontinuous strut to carry an axial compressive load of 180 kN. The length of strut between centre to centre of intersection is 2.5 m. The rivets of 18 mm diameter are provided at each end. Also design riveted connection.	(L3/CO3)	6
OR		
B) Design a steel beam for restrained against rotation at compression flange of clear span of 6 m resting on 230 mm support. A beam carries a DL + LL of 60 kN/m. Assume suitable data. Take $f_y = 250\text{MPa}$.	(L3/CO3)	12
Q. 3 Solve Any Two of the following.		12
A) List and explain components of crane girder.	Remember	6
B) Calculate the dead load per panel point for a simple Pratt truss using following data i) Span of truss = 24 m, ii) rise of truss = 4 m, iii) spacing of truss = 5 m, iv) total load = 2 kN/m ² .	(L3/CO1)	6
C) Calculate the live load per panel point for a simple Pratt truss using data given in Q.3 B considering live load = 1 kN/m ² .	(L3/CO1)	6

Q.4 Solve Any Two of the following.		12
A) Explain following concept with clear and labeled diagram:	Remember	6
a) Slab Base		
b) Gusseted Base		
c) Moment Resisting Base		
B) Calculate the safe compressive strength of a bridge compressive member consist of 2 channels ISMC-400 place toe to toe effective length of member is 7m the width over the back of channel 400 mm and the selection is properly connected by bracing.	(L3/CO3)	6
C) Design the base plate for a column made of ISHB 300 @ 63.0 kg/m to carry a compressive load of 1000 kN. The permissible stress of concrete is 3 MPa. The bearing capacity of soil is 250 kN/m ² . Assume $f_y = 250$ MPa.	(L3/CO3)	6
Q. 5 Solve Any Two of the following.		12
A) Define plastic hinge. Derive the equation to find length of plastic hinge for a simply supported rectangular beam of span 'L' subjected to a gradually increasing point load 'W' at the mid span.	(L3/CO4)	6
B) Define shape factor and calculate shape factor for	(L3/CO4)	6
i) rectangular section of 'b x d'		
ii) circular section of dia. 'd'.		
C) Differentiate between Plastic Design and Elastic Design.	Remember	6

***** End *****